

Rahim Iqbal

+94-75-9257307

rahimi.23@cse.mrt.ac.lk

Rahim-Iqbal

Rahim-Iqbal

Portfolio

Gampaha

Computer Science and Engineering undergraduate at the University of Moratuwa, specialising in Data Science. Experienced in applied machine learning, NLP, and building full-stack systems that hold up outside a demo.

Education

University of Moratuwa

2024 – Present

B.Sc. (Hons) Engineering – Computer Science Engineering, Data Science Stream (3rd Year)

Zahira College Colombo

2009 – 2023

G.C.E. Advanced Level – Physical Science Stream

- Distinctions in Combined Mathematics, Physics, and Chemistry

Research & Self Projects

Environmental Drivers of Respiratory Disease: A District Level

Analysis [arXiv:2607.04416](#) [Dataset](#)

Best Paper Award Nominee · Accepted, MERCon 2026, Moratuwa Engineering Research Conference Jan 2026 – May 2026

- Curated an 11-year district-level data lake spanning all 25 Sri Lankan districts by integrating satellite, pollutant, and health-admission sources for enterprise-grade analysis.
- Engineered temporally validated XGBoost predictive modeling for respiratory admission rate ($R^2 = 0.937$) and monthly PM_{2.5} ($R^2 = 0.976$), using SHAP for explainability and causal inference-style interpretation of air quality burden (80.1

Audiolit – *Interactive multimodal Explainable-AI workbench for speech recognition, emotion analytics, and deepfake detection* Jul 2026 – Oct 2026

an XAI engineering platform extending the ECHO 1.0 baseline to run ASR, speech emotion recognition, and audio deepfake detection with acoustic profiling and accent-bias auditing.

- Implemented an asynchronous Celery/RQ data pipeline for non-blocking inference, a deterministic SHA-256 content-addressed cache, and visual explainability methods including Grad-CAM and Integrated Gradients.
- Delivered the React web client with canvas-driven spectrogram mutation, real-time WebSocket monitoring, and interactive 2D/3D latent-space projections for engineering review.
- **Tools:** React, TypeScript, FastAPI, PyTorch, Hugging Face, Captum, Celery, Redis, MongoDB, Librosa, Docker

DiffDetect-YOLO

April 2025 – May 2025

- Developed a computer vision engineering pipeline to automate detection of visual differences between paired images for a Kaggle competition.
- Applied YOLOv8-World for open-vocabulary zero-shot detection, localising anomalies and generating bounding boxes for altered regions with strong classification performance.

GoldNest [Live](#) – *Micro-investment fintech platform, web and mobile*

March 2025

- Built a micro-investing platform for fractional gold from Rs. 100, adding wallets, automated savings plans, and physical redemption for end users.

- Tokenised holdings on Polygon for verifiable ownership, then layered AI-driven market analysis and gamification to improve user engagement.
- Shipped the companion iOS and Android client with biometric login, live pricing charts, and instant buy/sell flows.
- **Won 1st place at Dev{thon} 2.0** and reached the Startup Spark 2.0 finals with team Code Mavericks.
- **Tools:** Next.js, React, Node.js, Express, MongoDB, OpenZeppelin, Polygon, Groq SDK, Chart.js, JWT

MediBridge – Interoperable digital health passport

February 2025 – April 2025

- Engineered the backend entirely in Ballerina so patients can consolidate healthcare records across institutions and grant providers scoped access.
- Separated records, consent, and assist into independent services behind a central gateway with role-based access control and cleaner architectural boundaries.
- **Tools:** Ballerina, REST APIs, Docker

Skills

Machine Learning	PyTorch, scikit-learn, XGBoost, SHAP, Pandas, NumPy, Jupyter, YOLO
Frontend	React, Next.js, Tailwind CSS, Figma
Backend	FastAPI, Flask, REST APIs, Microservices, Apache Kafka
Database	PostgreSQL, MySQL
DevOps	Docker, Kubernetes, GitHub Actions, AWS
Languages	Python, TypeScript, JavaScript, SQL, C++, Ballerina
Tools	VS Code, PyCharm, CLion, Chrome DevTools, Hugging Face
Spoken Languages	English, Sinhala, Tamil

Achievements & Awards

1st Place, Dev{thon} 2.0 2025	Built GoldNest with team Code Mavericks, recognised for innovation in the final round. JISC eduCLaaS & Rotaract Club, University of Moratuwa.
2nd Runner-Up, OctWave 2.0 2025	Placed third among the top AI/ML teams from Sri Lankan universities, across machine learning and computer vision. Competition partnered with Tortil Inc. IEEE IAS Student Branch Chapter, University of Moratuwa.
Finalist, Startup Spark 2.0, Techno 2025	Pitched GoldNest with team Code Mavericks at the “Innovate for Impact” startup competition, held at the BMICH, Colombo. IESL Student Chapter, University of Moratuwa.
Finalist, AIthon 2026 2026	Advanced to the MVP Development Phase with team MatriXplorers, out of 60+ teams. Enterprise AI Innovation Challenge, Hemas Holdings PLC.

Extra Curricular Activities

Youth Initiative Circle	Active volunteer and mentor in a grassroots community initiative dedicated to supporting my home village. Facilitate educational guidance and skill-building activities to empower local youth and foster community development.
Majlis Ul Islam	Assisted Majlis ul Islam with first-year student mentorship, Ramadan logistics, and social media outreach campaigns.

Certifications

Microservices and CI/CD Pipeline Builder

AWS Academy, Jan 2026

Cloud Foundations

AWS Academy, Jan 2026

Generative AI Mastermind

Outskills, 2026

Referees

Prof. Uthayasanker Thayasivam

Former Head of the Department, Senior Lecturer
Department of Computer Science and Engineering
University of Moratuwa, Sri Lanka

rtuthaya@cse.mrt.ac.lk

Prof. Dulani Meedeniya

Senior Lecturer
Department of Computer Science and Engineering
University of Moratuwa, Sri Lanka

dulanim@cse.mrt.ac.lk